

MAD & PWA LAB

Experiment 5

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Aim: To apply navigation, routing and gestures in Flutter App

Theory:

1. Navigation in Flutter

Navigation allows users to move between screens in an app. It is managed using the Navigator and a stack-based system. The most common navigation actions include:

- Push Navigation: Moves to a new screen.
- Pop Navigation: Goes back to the previous screen.
- Named Routes: Uses predefined names for navigation instead of directly creating new screen instances.

2. Routing in Flutter

Routing defines the path and logic for screen transitions. There are two main types:

- Static Routing: Predefined routes set in MaterialApp.
- Dynamic Routing: Routes created dynamically using onGenerateRoute, allowing flexible screen navigation.

3. Gestures in Flutter

Gestures enable user interactions such as tapping, swiping, and dragging. Flutter provides:

- GestureDetector: Detects custom gestures like tap, double-tap, and long press.
- InkWell: Adds ripple effects for a better touch response.

Code: `elcome/widgets/privacy_and_terms.dart';`

Landing page:

```
import 'package:flutter/material.dart';

import
'package:whatsapp_messenger/common/
routes/routes.dart';

import
'package:whatsapp_messenger/common/
utils/coloors.dart';

import
'package:whatsapp_messenger/common/
widgets/custom_elevated_button.dart';

import
'package:whatsapp_messenger/feature/w
elcome/widgets/language_button.dart';

import
'package:whatsapp_messenger/feature/w

class WelcomePage extends
StatelessWidget {

  const WelcomePage({super.key});

  navigateToLoginPage(context) {
    Navigator.of(context)
      .pushNamedAndRemoveUntil(Routes.
login, (route) => false);
  }

  @override
  Widget build(BuildContext context) {
```

```
return Scaffold(  
  body: Column(  
    children: [  
      Expanded(  
        child: Align(  
          alignment:  
Alignment.bottomCenter,  
        child: Padding(  
          padding:  
            const  
EdgeInsets.symmetric(horizontal: 50,  
vertical: 10),  
          child:  
Image.asset('assets/images/circle.png',  
            color: Colors.greenDark),  
        ),  
      ),  
      const SizedBox(height:  
40), Expanded(  
        child: Column(  
          children: [  
            Expanded(  
              child: Image.asset('assets/images/circle.png',  
                color: Colors.greenDark),  
            ),  
            const SizedBox(height: 40),  
            Expanded(  
              child: Image.asset('assets/images/circle.png',  
                color: Colors.greenDark),  
            ),  
          ],  
        ),  
      ),  
    ],  
  ),  
);
```

```

children: [ const Text(
    'Welcome to WhatsApp',
    style: TextStyle(
      fontSize: 22,
      fontWeight: FontWeight.bold,
    ),
  ),
  const PrivacyAndTerms(),
  CustomElevatedButton(
    onPressed: () =>
navigateToLoginPage(context),
    text: 'AGREE AND CONTINUE',
  ),
  const SizedBox(height: 50),
  LanguageButton(),
],
),
),
],
),
);
}
}

```

Authentication page:

```

import 'package:flutter/material.dart';

import
'package:flutter_riverpod/flutter_riverpod
.dart';

```

```

import
'package:whatsapp_messenger/common/
extension/custom_theme_extension.dart';

import
'package:whatsapp_messenger/feature/a
uth/controller/auth_controller.dart';

import
'package:whatsapp_messenger/feature/a
uth/widgets/custom_text_field.dart';

class VerificationPage extends
ConsumerWidget {
  const VerificationPage({
    super.key,
    required this.smsCodeId,
    required this.phoneNumber,
  });

  final String smsCodeId;
  final String phoneNumber;

  Future<void> verifySmsCode(
    BuildContext context,
    WidgetRef ref,
    String smsCode,
  ) async {
    await
ref.read(authControllerProvider).verifySms
Code(
    context: context,
    smsCodeId: smsCodeId,
    smsCode: smsCode,

```

```

    );
}

@override
Widget build(BuildContext context,
WidgetRef ref) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor:
Theme.of(context).scaffoldBackgroundCol
or,
      elevation: 0,
      centerTitle: true,
      title: Text(
        'Verify your number',
        style: TextStyle(
          color:
context.theme.authAppBarTextColor,
        ),
      ),
      actions: [
        IconButton(
          onPressed: () {},
          icon: const Icon(Icons.more_vert),
        ),
      ],
    ),
    body: SingleChildScrollView(
      padding: const
EdgeInsets.symmetric(horizontal: 20),
      child: Column(

```

```

        children: [
          Padding(
            padding: const
EdgeInsets.symmetric(horizontal: 10),
            child: RichText(
              textAlign: TextAlign.center,
              text: TextSpan(
                style: TextStyle(color:
context.theme.greyColor),
                children: [
                  const TextSpan(
                    text:
                      "You've tried to register
before requesting an SMS or Call with your
code.",
                  ),
                  TextSpan(
                    text: " Wrong number?",
                    style: TextStyle(
                      color:
context.theme.blueColor,
                    ),
                  ),
                ],
              ),
            ),
          ),
          const SizedBox(height: 20),
          Container(
            padding: const
EdgeInsets.symmetric(horizontal: 80),

```

```

        child: CustomTextField(
            hintText: " ----- ",
            fontSize: 30,
            autoFocus: true,
            keyboardType:
TextInputType.number,
            onChanged: (value) async {
                if (value.length == 6) {
                    await verifySmsCode(context,
ref, value);
                }
            },
        ),
    ),
    const SizedBox(height: 20),
    Text(
        'Enter 6-digit code',
        style: TextStyle(color:
context.theme.greyColor),
    ),
    const SizedBox(height: 30),
    Row(
        children: [
            Icon(Icons.message, color:
context.theme.greyColor),
            const SizedBox(width: 20),
            Text(
                'Resend SMS',
                style: TextStyle(
                    color: context.theme.greyColor,
                ),
            ),
        ],
    ),
    const SizedBox(height: 10),
    Divider(
        color:
context.theme.greyColor!.withOpacity(.2),
    ),
    const SizedBox(height: 10),
    Row(
        children: [
            Icon(Icons.phone, color:
context.theme.greyColor),
            const SizedBox(width: 20),
            Text(
                'Call Me',
                style: TextStyle(
                    color: context.theme.greyColor,
                ),
            ),
        ],
    ),
    const SizedBox(width: 20),
    Text(
        'Call Me',
        style: TextStyle(
            color: context.theme.greyColor,
        ),
    ),
    ],
),
);
}
}

```

Chats page:

```
import
'package:cached_network_image/cached_network_image.dart';

import 'package:flutter/material.dart';

import
'package:flutter_riverpod/flutter_riverpod.dart';

import 'package:intl/intl.dart';

import
'package:whatsapp_messenger/common/extension/custom_theme_extension.dart';

import
'package:whatsapp_messenger/common/models/last_message_model.dart';

import
'package:whatsapp_messenger/common/models/user_model.dart';

import
'package:whatsapp_messenger/common/routes/routes.dart';

import
'package:whatsapp_messenger/common/utils/colors.dart';

import
'package:whatsapp_messenger/feature/chat/controller/chat_controller.dart';

class ChatHomePage extends ConsumerWidget {
  const ChatHomePage({super.key});

  List<LastMessageModel>
  getDemoChats() {
    return [
```

```
      LastMessageModel(
        username: 'Dev Stack',
        contactId: '1',
        profileImageUrl:
        'https://picsum.photos/seed/1/200',
        lastMessage: 'Hi Everyone 🦄',
        timeSent:
        DateTime.now().subtract(const Duration(minutes: 4)),
      ),
      LastMessageModel(
        username: 'Kishor',
        contactId: '2',
        profileImageUrl:
        'https://picsum.photos/seed/2/200',
        lastMessage: 'Hi Kishor',
        timeSent:
        DateTime.now().subtract(const Duration(hours: 1)),
      ),
      LastMessageModel(
        username: 'Dev Server Chat',
        contactId: '3',
        profileImageUrl:
        'https://picsum.photos/seed/3/200',
        lastMessage: 'Alice: The new API is ready for testing',
        timeSent:
        DateTime.now().subtract(const Duration(hours: 2)),
      ),
      LastMessageModel(
```

```

        username: 'Collins',
        contactId: '4',
        imageUrl:
'https://picsum.photos/seed/4/200',
        lastMessage: 'Hey Dev Stack! How\'s
the project going?',
        timeSent:
DateTime.now().subtract(const
Duration(hours: 4)),
    ),
    LastMessageModel(
        username: 'Friends',
        contactId: '5',
        imageUrl:
'https://picsum.photos/seed/5/200',
        lastMessage: 'John: Anyone up for
coffee? 🐼',
        timeSent:
DateTime.now().subtract(const
Duration(hours: 5)),
    ),
    LastMessageModel(
        username: 'Flutter Community',
        contactId: '6',
        imageUrl:
'https://picsum.photos/seed/6/200',
        lastMessage: 'New flutter 3.0 features
announced! 🌈',
        timeSent:
DateTime.now().subtract(const
Duration(hours: 10)),
    ),
    LastMessageModel(

```

```

        username: 'Balram Rathore',
        contactId: '7',
        imageUrl:
'https://picsum.photos/seed/7/200',
        lastMessage: 'happy coding
🟡 🟡 🟡',
        timeSent:
DateTime.now().subtract(const
Duration(hours: 10)),
    ),
    ];
}

@override
Widget build(BuildContext context,
WidgetRef ref) {
    return Scaffold(
        backgroundColor: const
Color(0xFF121B22),
        body: ListView.builder(
            itemCount: getDemoChats().length,
            shrinkWrap: true,
            itemBuilder: (context, index) {
                final lastMessageData =
getDemoChats()[index];

                return Column(
                    children: [
                        ListTile(
                            onTap: () {
                                Navigator.pushNamed(

```

```

        context,
        Routes.chat,
        arguments: UserModel(
            username:
lastMessageData.username,
            uid:
lastMessageData.contactId,
            profileImageUrl:
lastMessageData.profileImageUrl,
            active: true,
            lastSeen: 0,
            phoneNumber: '0',
            groupId: [],
        ),
    );
},
title: Row(
    mainAxisAlignment:
MainAxisAlignment.spaceBetween,
    children: [
        Expanded(
            child: Text(
                lastMessageData.username,
                style: const TextStyle(
                    fontWeight:
FontWeight.w500,
                    color: Colors.white,
                ),
                overflow:
TextOverflow.ellipsis,
            ),

```

```

        ),
        Text(
            DateFormat.Hm().format(last
MessageData.timeSent),
            style: const TextStyle(
                fontSize: 13,
                color: Colors.grey,
            ),
        ),
    ],
),
    subtitle: Padding(
        padding: const
EdgeInsets.only(top: 3),
        child: Row(
            children: [
                const Icon(
                    Icons.done_all,
                    size: 16,
                    color: Colors.greenDark,
                ),
                const SizedBox(width: 3),
                Expanded(
                    child: Text(
                        lastMessageData.lastMessa
ge,
                        maxLines: 1,
                        overflow:
TextOverflow.ellipsis,
                        style: const TextStyle(
                            color: Colors.grey,

```



```

        ),
        ),
        ),
    ],
    ),
    ),
    leading: CircleAvatar(
        backgroundImage:
CachedNetworkImageProvider(
        lastMessageData.profileImage
Url,
    ),
    radius: 24,
    ),
    ),
    Divider(
        color: Colors.grey[900],
        indent: 80,
        height: 0,
    ),
    ],
    );
},
),
floatingActionButton:
FloatingActionButton(
    backgroundColor: Colors.greenDark,
    onPressed: () {
        Navigator.pushNamed(context,
Routes.contact);
    },

```

```

        child: const Icon(
            Icons.message,
            color: Colors.white,
        ),
    ),
    );
}
}

```

Status Page:

```

import 'dart:html' as html;
import 'package:flutter/material.dart';
import
'package:image_picker/image_picker.dart'
;

```

// Enhanced Status Model

```

class Status {
    final String id;
    final String name;
    final String imageUrl; // Changed from
imagePath to imageUrl
    final DateTime timestamp;
    final bool isViewed;
    final StatusType type;
    final String? caption;

    Status({
        required this.id,

```

```

    required this.name,
    required this.imageUrl,
    required this.timestamp,
    this.isViewed = false,
    this.type = StatusType.image,
    this.caption,
  });
}

enum StatusType { image, text }

class StatusHomePage extends
StatefulWidget {
  const StatusHomePage({super.key});

  @override
  State<StatusHomePage> createState() =>
    _StatusHomePageState();
}

class _StatusHomePageState extends
State<StatusHomePage> {
  final ImagePicker _picker = ImagePicker();
  List<Status> recentUpdates = [];
  List<Status> viewedUpdates = [];
  Status? myCurrentStatus;

  @override
  void initState() {
    super.initState();

    _loadDemoStatuses();
  }

  void _loadDemoStatuses() {
    // Load demo statuses with web-
    compatible URLs
    recentUpdates = [
      Status(
        id: '1',
        name: 'Balram Rathore',
        imageUrl:
        'https://picsum.photos/400/600', // Using
        placeholder image
        timestamp: DateTime.now(),
      ),
      Status(
        id: '2',
        name: 'Kishor Kumar',
        imageUrl:
        'https://picsum.photos/400/601', //
        Different placeholder
        timestamp: DateTime.now(),
      ),
    ];

    viewedUpdates = [
      Status(
        id: '3',
        name: 'Saket Sinha',
        imageUrl:
        'https://picsum.photos/400/602',

```

```

        timestamp:
DateTime.now().subtract(const
Duration(hours: 1)),
        isViewed: true,
    ),
];
}

Future<void> _addImageStatus() async {
    try {
        final XFile? image = await
_picker.pickImage(source:
ImageSource.gallery);

        if (image == null) return;

        // Create object URL for web

        final bytes = await
image.readAsBytes();

        final blob = html.Blob([bytes]);

        final url =
html.Url.createObjectUrlFromBlob(blob);

        setState(() {
            myCurrentStatus = Status(
                id: DateTime.now().toString(),
                name: 'My Status',
                imageUrl: url,
                timestamp: DateTime.now(),
                type: StatusType.image,
            );

```

```

        // Add to recent updates
        recentUpdates.insert(0,
myCurrentStatus!);
    });
} catch (e) {
    showDialog(
        context: context,
        builder: (context) => AlertDialog(
            title: const Text('Error'),
            content: Text('Failed to add status:
$e'),
            actions: [
                TextButton(
                    onPressed: () =>
Navigator.pop(context),
                    child: const Text('OK'),
                ),
            ],
        ),
    );
}
}

Future<void> _addTextStatus() async {
    final TextEditingController textController
= TextEditingController();

    final result = await showDialog<String>(
        context: context,
        builder: (context) => AlertDialog(
            backgroundColor: const
Color(0xFF121B22),

```

```

        title: const Text('Add Text Status',
          style: TextStyle(color:
Colors.white)),
        content: TextField(
          controller: textController,
          style: const TextStyle(color:
Colors.white),
          maxLines: 3,
          decoration: const InputDecoration(
            hintText: 'Enter your status...',
            hintStyle: TextStyle(color:
Colors.grey),
            border: OutlineInputBorder(),
          ),
        ),
        actions: [
          TextButton(
            onPressed: () =>
Navigator.pop(context),
            child: const Text('Cancel'),
          ),
          TextButton(
            onPressed: () =>
Navigator.pop(context,
textController.text),
            child: const Text('Post'),
          ),
        ],
      ),
    );

```

```

    if (result != null && result.isNotEmpty) {
      setState(() {
        myCurrentStatus = Status(
          id: DateTime.now().toString(),
          name: 'My Status',
          imageUrl: '', // Empty for text status
          timestamp: DateTime.now(),
          type: StatusType.text,
          caption: result,
        );

        // Add to recent updates
        recentUpdates.insert(0,
myCurrentStatus!);
      });
    }
  }

  void _viewStatus(Status status) {
    Navigator.push(
      context,
      MaterialPageRoute(
        builder: (context) =>
StatusViewPage(status: status),
      ),
    );

    // Mark status as viewed
    if (!status.isViewed &&
recentUpdates.contains(status)) {

```

```

    setState(() {
      final index =
recentUpdates.indexOf(status);

      final viewedStatus = Status(
        id: status.id,
        name: status.name,
        imageUrl: status.imageUrl,
        timestamp: status.timestamp,
        isViewed: true,
        type: status.type,
        caption: status.caption,
      );

      recentUpdates.removeAt(index);
      viewedUpdates.insert(0,
viewedStatus);
    });
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: const
Color(0xFF121B22),
    body: SingleChildScrollView(
      child: Column(
        crossAxisAlignment:
CrossAxisAlignment.start,
        children: [
          // My Status Section

```

```

ListTile(
  leading: Stack(
    children: [
      CircleAvatar(
        radius: 30,
        backgroundColor: Colors.grey,
        backgroundImage:
myCurrentStatus?.type ==
StatusType.image
          ?
NetworkImage(myCurrentStatus!.imageUr
l)
          : null,
        child: myCurrentStatus == null
          ? const Icon(Icons.person,
            color: Colors.white, size:
35)
          : myCurrentStatus?.type ==
StatusType.text
            ? const
Icon(Icons.text_fields, color: Colors.white)
            : null,
      ),
      Positioned(
        bottom: 0,
        right: 0,
        child: Container(
          padding: const
EdgeInsets.all(4),
          decoration: BoxDecoration(
            color: Colors.green,

```

```

        borderRadius:
BorderRadius.circular(12),
    ),
    child: const Icon(
      Icons.add,
      color: Colors.white,
      size: 20,
    ),
  ),
],
),
title: const Text(
  'My status',
  style: TextStyle(
    color: Colors.white,
    fontWeight: FontWeight.bold,
  ),
),
subtitle: Text(
  myCurrentStatus != null
    ? 'Tap to view your status'
    : 'Tap to add status update',
  style: const TextStyle(color:
Colors.grey),
),
onTap: () {
  if (myCurrentStatus != null) {
    _viewStatus(myCurrentStatus!);
  }
}
      },
    ),
  ),
  if (recentUpdates.isNotEmpty) ...[
    const Padding(
      padding:
EdgeInsets.symmetric(horizontal: 15,
vertical: 8),
      child: Text(
        'Recent updates',
        style: TextStyle(
          color: Colors.grey,
          fontWeight: FontWeight.bold,
        ),
      ),
    ),
    ListView.builder(
      shrinkWrap: true,
      physics: const
NeverScrollableScrollPhysics(),
      itemCount:
recentUpdates.length,
      itemBuilder: (context, index) =>
StatusTile(
        status: recentUpdates[index],
        onTap: () =>
_viewStatus(recentUpdates[index]),
      ),
    ),
  ],
]

```

```

    if (viewedUpdates.isNotEmpty) ...[
      const Padding(
        padding:
EdgeInsets.symmetric(horizontal: 15,
vertical: 8),
        child: Text(
          'Viewed updates',
          style: TextStyle(
            color: Colors.grey,
            fontWeight: FontWeight.bold,
          ),
        ),
      ),
    ],
    ListView.builder(
      shrinkWrap: true,
      physics: const
NeverScrollableScrollPhysics(),
      itemCount:
viewedUpdates.length,
      itemBuilder: (context, index) =>
StatusTile(
        status: viewedUpdates[index],
        onTap: () =>
_viewStatus(viewedUpdates[index]),
      ),
    ],
  ],
),
),
floatingActionButton: Column(

```

```

      mainAxisAlignment:
MainAxisAlignment.end,
      children: [
        FloatingActionButton(
          mini: true,
          onPressed: _addTextStatus,
          backgroundColor: Colors.grey[700],
          child: const Icon(Icons.edit),
        ),
        const SizedBox(height: 16),
        FloatingActionButton(
          onPressed: _addImageStatus,
          backgroundColor: Colors.teal[700],
          child: const Icon(Icons.camera_alt),
        ),
      ],
    ),
  );
}
}

```

```

class StatusTile extends StatelessWidget {
  final Status status;
  final VoidCallback onTap;

  const StatusTile({
    super.key,
    required this.status,
    required this.onTap,
  });
}

```

```

@override
Widget build(BuildContext context) {
  return ListTile(
    leading: Container(
      decoration: BoxDecoration(
        shape: BoxShape.circle,
        border: Border.all(
          color: status.isViewed ? Colors.grey
: Colors.green,
          width: 2,
        ),
      ),
    child: CircleAvatar(
      radius: 28,
      backgroundColor: Colors.grey,
      backgroundImage: status.type ==
StatusType.image
        ? NetworkImage(status.imageUrl)
        : null,
    child: status.type == StatusType.text
        ? const Icon(Icons.text_fields,
color: Colors.white)
        : null,
    ),
  ),
  title: Text(
    status.name,
    style: const TextStyle(
      color: Colors.white,

```

```

fontWeight: FontWeight.bold,
    ),
  ),
  subtitle: Text(
    'Today,
    ${status.timestamp.hour}:${status.timesta
mp.minute.toString().padLeft(2, '0')}',
    style: const TextStyle(color:
Colors.grey),
  ),
  onTap: onTap,
);
}
}

```

```

class StatusViewPage extends
StatelessWidget {

```

```

  final Status status;

```

```

  const StatusViewPage({super.key,
required this.status});

```

```

  @override

```

```

Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: Colors.black,
    appBar: AppBar(
      backgroundColor: Colors.transparent,
      title: Text(status.name),
      iconTheme: const
IconThemeData(color: Colors.white),

```



```

    ),
    body: Center(
      child: status.type == StatusType.image
        ? Image.network(
            status.imageUrl,
            fit: BoxFit.contain,
            loadingBuilder: (context, child,
loadingProgress) {
              if (loadingProgress == null)
return child;

              return const
CircularProgressIndicator();
            },
            errorBuilder: (context, error,
stackTrace) {
              return const Icon(
                Icons.error,
                color: Colors.red,
                size: 48,
              );
            },
        )
      : Container(
          padding: const EdgeInsets.all(20),
          color: Colors.teal,
          child: Center(
            child: Text(
              status.caption ?? "",
              style: const TextStyle(
                color: Colors.white,
                fontSize: 24,

```

```

            ),
            textAlign: TextAlign.center,
          ),
        ),
      ),
    );
  }
}

```

Call page:

```
import 'package:flutter/material.dart';
```

```

class Call {
  final String name;
  final String avatarUrl;
  final DateTime timestamp;
  final bool isIncoming;
  final bool isMissed;
  final bool isVideo;

  Call({
    required this.name,
    required this.avatarUrl,
    required this.timestamp,
    this.isIncoming = true,
    this.isMissed = false,
    this.isVideo = false,
  });
}

```

```

class CallHomePage extends StatefulWidget {
  const CallHomePage({super.key});

  @override
  State<CallHomePage> createState() =>
    _CallHomePageState();
}

```

```

class _CallHomePageState extends
State<CallHomePage> {

  // Expanded demo call data with Picsum
  images

  final List<Call> calls = [

    Call(
      name: "Mike Johnson",
      avatarUrl:
        "https://picsum.photos/seed/mike/200",
      timestamp: DateTime.now().subtract(const
        Duration(minutes: 5)),
      isIncoming: true,
      isVideo: false,
    ),
    Call(
      name: "Sarah Wilson",
      avatarUrl:
        "https://picsum.photos/seed/sarah/200",
      timestamp: DateTime.now().subtract(const
        Duration(minutes: 15)),
      isIncoming: false,
      isVideo: true,
    ),
    Call(

```

```

      name: "Alex Thompson",
      avatarUrl:
        "https://picsum.photos/seed/alex/200",
      timestamp: DateTime.now().subtract(const
        Duration(hours: 2)),
      isIncoming: true,
      isMissed: true,
      isVideo: false,
    ),
    Call(
      name: "Emma Davis",
      avatarUrl:
        "https://picsum.photos/seed/emma/200",
      timestamp: DateTime.now().subtract(const
        Duration(hours: 3)),
      isIncoming: false,
      isVideo: true,
    ),
    Call(
      name: "James Miller",
      avatarUrl:
        "https://picsum.photos/seed/james/200",
      timestamp: DateTime.now().subtract(const
        Duration(hours: 5)),
      isIncoming: true,
      isMissed: false,
      isVideo: true,
    ),
    Call(
      name: "Lisa Brown",
      avatarUrl:
        "https://picsum.photos/seed/lisa/200",
      timestamp: DateTime.now().subtract(const
        Duration(hours: 6)),

```

```

        isIncoming: false,

        isMissed: true,

        isVideo: false,
    ),
    Call(
        name: "David Clark",

        avatarUrl:
"https://picsum.photos/seed/david/200",

        timestamp: DateTime.now().subtract(const
Duration(hours: 8)),

        isIncoming: true,

        isVideo: false,
    ),
    Call(
        name: "Rachel Green",

        avatarUrl:
"https://picsum.photos/seed/rachel/200",

        timestamp: DateTime.now().subtract(const
Duration(days: 1)),

        isIncoming: false,

        isVideo: true,
    ),
    Call(
        name: "Tom Wilson",

        avatarUrl:
"https://picsum.photos/seed/tom/200",

        timestamp: DateTime.now().subtract(const
Duration(days: 1, hours: 3)),

        isIncoming: true,

        isMissed: true,

        isVideo: false,
    ),
    Call(

```

```

        name: "Emily White",

        avatarUrl:
"https://picsum.photos/seed/emily/200",

        timestamp: DateTime.now().subtract(const
Duration(days: 1, hours: 6)),

        isIncoming: false,

        isVideo: true,
    ),
];

```

```

void _showOutgoingCallDialog(
    BuildContext context, String name, bool
isVideo) {
    showDialog(
        context: context,
        barrierDismissible: false,
        builder: (context) => Align(
            alignment: Alignment.topCenter,
            child: OutgoingCallDialog(name: name,
isVideo: isVideo),
        ),
    );
}

```

```

String _getTimeString(DateTime timestamp) {
    final now = DateTime.now();

    final difference =
now.difference(timestamp);

    if (difference.inDays > 0) {
        return "Yesterday,
${timestamp.hour.toString().padLeft(2,
'0')}:${timestamp.minute.toString().padLeft(2,
'0')}";
    }
}

```

```

    } else if (difference.inHours > 0) {
        return "${difference.inHours}h ago";
    } else {
        return
"$${timestamp.hour.toString().padLeft(2,
'0')}:${timestamp.minute.toString().padLeft(2,
'0')}";
    }
}

```

```

@override
Widget build(BuildContext context) {
    return Scaffold(
        backgroundColor: const
Color(0xFF1F2C34),
        body: SafeArea(
            child: Column(
                children: [
                    Expanded(
                        child: ListView.builder(
                            itemCount: calls.length,
                            itemBuilder: (context, index) {
                                final call = calls[index];
                                return ListTile(
                                    leading: CircleAvatar(
                                        radius: 25,
                                        backgroundImage:
NetworkImage(call.avatarUrl),
                                        backgroundColor:
Colors.grey[700],
                                    ),
                                    title: Text(
                                        call.name,

```

```

style: const TextStyle(
    color: Colors.white,
    fontWeight: FontWeight.w500,
),
),
subtitle: Row(
    children: [
        Icon(
            call.isIncoming
                ? Icons.call_received
                : Icons.call_made,
            size: 15,
            color: call.isMissed ? Colors.red
: Colors.green,
        ),
        const SizedBox(width: 4),
        Text(
            _getTimeString(call.timestamp),
            style: TextStyle(
                color: Colors.grey[400],
                fontSize: 13,
            ),
        ),
    ],
),
trailing: IconButton(
    icon: Icon(
        call.isVideo ? Icons.videocam :
Icons.call,
        color: const Color(0xFF00A884),
    ),
    onPressed: () {

```



```

        color: Colors.white,
      },
      fontSize: 16,
    },
    fontWeight: FontWeight.bold,
  ),
),
Text(
  "Calling${isVideo ? ' video' : ''}...",
  style: TextStyle(
    color: Colors.grey[400],
    fontSize: 14,
  ),
),
],
),
),
IconButton(
  icon: const Icon(Icons.close, color:
Colors.red),
  onPressed: () =>
Navigator.of(context).pop(),
),
IconButton(
  icon: Icon(
    isVideo ? Icons.videocam : Icons.call,
    color: Colors.green,
  ),
  onPressed: () =>
Navigator.of(context).pop(),
),
],
),
);

```

Routing Code:

```
import 'package:flutter/material.dart';
import 'package:page_transition/page_transition.dart';
import 'package:whatsapp_messenger/common/models/user_model.dart';
import 'package:whatsapp_messenger/feature/auth/pages/login_page.dart';
import 'package:whatsapp_messenger/feature/auth/pages/user_info_page.dart';
import 'package:whatsapp_messenger/feature/auth/pages/verification_page.dart';
import 'package:whatsapp_messenger/feature/chat/pages/chat_page.dart';
import 'package:whatsapp_messenger/feature/chat/pages/profile_page.dart';
import 'package:whatsapp_messenger/feature/contact/pages/contact_page.dart';
import 'package:whatsapp_messenger/feature/home/pages/home_page.dart';
import 'package:whatsapp_messenger/feature/welcome/pages/welcome_page.dart';
```

```
class Routes {
  static const String welcome = 'welcome';
  static const String login = 'login';
  static const String verification = 'verification';
  static const String userInfo = 'user-info';
  static const String home = 'home';
  static const String contact = 'contact';
  static const String chat = 'chat';
  static const String profile = 'profile';

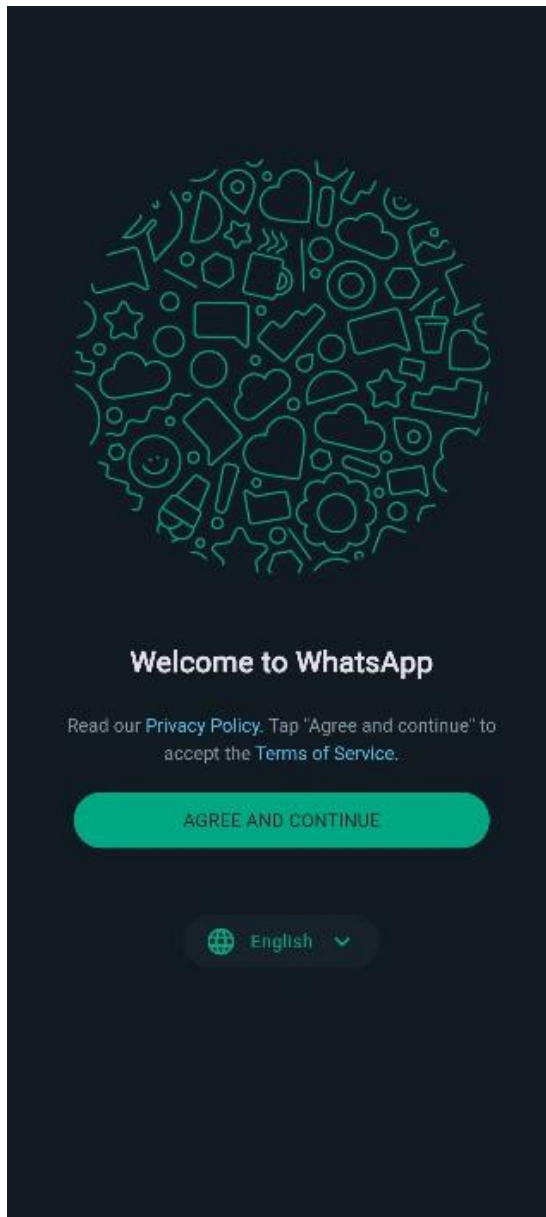
  static Route<dynamic> onGenerateRoute(RouteSettings settings) {
    switch (settings.name) {
      case welcome:
        return PageTransition(
          child: const WelcomePage(),
          type: PageTransitionType.rightToLeft,
          duration: const Duration(milliseconds: 500),
        );
      case login:
        return PageTransition(
          child: const LoginPage(),
          type: PageTransitionType.rightToLeft,
          duration: const Duration(milliseconds: 500),
        );
      case verification:
        final args = settings.arguments as Map<String, dynamic>;
        if (args == null ||
            !args.containsKey('smsCodeId') ||
            !args.containsKey('phoneNumber')) {
          throw ArgumentError("Invalid arguments for VerificationPage");
        }
        return PageTransition(
          child: VerificationPage(
            smsCodeId: args['smsCodeId'],
            phoneNumber: args['phoneNumber'],
          ),
          type: PageTransitionType.fade,
          duration: const Duration(milliseconds: 600),
        );
      case userInfo:
        final String? profileImageUrl = settings.arguments as String?;
```

```

    return PageTransition(
      child: UserInfoPage(profileImageUrl: profileImageUrl),
      type: PageTransitionType.rightToLeft,
      duration: const Duration(milliseconds: 500),
    );
  case home:
    return PageTransition(
      child: const HomePage(),
      type: PageTransitionType.fade,
      duration: const Duration(milliseconds: 600),
    );
  case contact:
    return PageTransition(
      child: const ContactPage(),
      type: PageTransitionType.rightToLeft,
      duration: const Duration(milliseconds: 500),
    );
  case chat:
    final user = settings.arguments as UserModel?;
    if (user == null) {
      throw ArgumentError("UserModel argument is required for ChatPage");
    }
    return PageTransition(
      child: ChatPage(user: user),
      type: PageTransitionType.rightToLeftWithFade,
      duration: const Duration(milliseconds: 600),
    );
  case profile:
    final user = settings.arguments as UserModel?;
    if (user == null) {
      throw ArgumentError("UserModel argument is required for ProfilePage");
    }
    return PageTransition(
      child: ProfilePage(user: user),
      type: PageTransitionType.fade,
      duration: const Duration(milliseconds: 800),
    );
  default:
    return MaterialPageRoute(
      builder: (context) => const Scaffold(
        body: Center(
          child: Text('No Page Route Provided'),
        ),
      ),
    );
}
}
}

```


Output:



Verify your number

You've tried to register before requesting an SMS or Call with your code. **Wrong number?**

Enter 6-digit code

Resend SMS

Call Me

Privacy - Terms

WhatsApp

CHATS

STATUS

CALLS

Dev Stack

Hi Everyone

01:47

Kishor

Hi Kishor

00:51

Dev Server Chat

Alice: The new API is ready for testing

23:51

Collins

Hey Dev Stack! How's the project going?

21:51

Friends

John: Anyone up for coffee?

20:51

Flutter Community

New flutter 3.0 features announced!

15:51

Balram Rathore

happy coding

15:51

